

Grubeni: 3500 Dm

Davanje jada

na 29.11.24

Dano:

$$T_x = 500K$$

$$T_n = 400K$$

$$m = 500Kl$$

$$q = 1,26 \cdot 10^2 \frac{Dm}{Kl}$$

Amer - ?

~10,5.
Jednake:

$$\eta = 1 - \frac{T_{max}}{T_n} = 1 - \frac{100}{400} = 0,75$$

$$Q = m \cdot q = 6,3 \cdot 10^9 Dm$$

$$A = \eta \cdot Q = 1,525 \cdot 10^9 Dm$$

$$Grubeni: A = 1,525 \cdot 10^9 Dm$$

Dano:

$$P = 14200 Bn$$

$$t = 9640 L$$

$$a = 1,1 Kl$$

$$q = 33 \cdot 10^6 \frac{Dm}{Kl}$$

$$T_1 = 473,15 K$$

$$T_2 = 331,15$$

$$\eta, \eta' = ?$$

$$\eta \approx 0,19$$

$$\eta' \approx 0,7$$

~10,19
Jednake:

$$A = P \cdot t = 14200 \cdot 9640 = 1,37 \cdot 10^6 Dm$$

$$Q = m \cdot q = 1,1 \cdot 33 \cdot 10^6 = 3,63 \cdot 10^6 Dm$$

$$\eta = \frac{A}{Q} = \frac{1,37 \cdot 10^6}{3,63 \cdot 10^6} \approx 0,19$$

$$\eta_{kaym} = 1 - \frac{T_2}{T_1} \approx 0,5002$$

$$\eta_{kaym} \approx 0,3002; \eta' \approx 0,7$$

Dano:

$$\eta = 0,4$$

$$P = 400 Dm$$

$$A = ?$$

$$A = ?$$

~10,27
Jednake:

$$\eta = 1 - \frac{T_2}{T_1}; 0,4 = 1 - \frac{T_2}{T_1}; \frac{T_2}{T_1} = 0,6$$

$$A = ?$$

$$A_{12} + A_{23} + A_{34} + A_{41} = 0$$

~~R.R. Hypothese $Q_{12} = 0$ (unphysikalisch)~~

$$A_{12} = (\eta - 1) A_{11}$$

$$= -0.6 \cdot 2000 = -1200 \text{ Dmc}$$

Ges.: $A_{12} = -1200 \text{ Dmc}$

Dopp.: $\eta = 10.22$
 Temperatur:
 $T_1 = 500 \text{ K}$
 $T_2 = 300 \text{ K}$
 $A = 2000 \text{ Dmc}$
 $\eta, Q = ?$

Denkweise:
 $\eta = 1 - \frac{T_2}{T_1} = 0.4$
 $A = Q_1 - Q_2$
 Typ ungenutzten Energie:
 $A = Q$
 $A_{12} = 2000 \text{ Dmc}$
 $A_{12} + A_{23} + A_{34} + A_{41} = 0$
 $A_{12} + A_{34} = 0$
 $A_{34} = -A_{12} = -2000 \text{ Dmc}$
 $Q_2 = A_{34} = 2000 \text{ Dmc}$

Ges.: $\eta = 0.4, Q_2 = 2000 \text{ Dmc}$

Demo:
 $T_2 =$
 $T_1 =$
 $m_1 =$
 $m_2 =$
 3) m
 $Q_2 =$
 Ges.
 D
 $V =$
 $V_{n, m}$
 P_{max}
 $AS =$
 2)